



HOW WE TESTED

Performance: Each iPod dock was subjected to a series of frequency tests. The first volley of test samples included tones at various frequencies such as 30 Hz, 50 Hz, 100 Hz, 250 Hz etc. This was followed by a 16bit 22kHz - 20Hz sweep. Here we checked for distortion and any sort of clipping. Scores were given on a scale of 1 to 10 but with a fixed performance threshold. That means that only when a dock was able to reproduce a tone at a certain quality level was it given an overall good score. This was done by amplifying the scores on 10 to scores on 100 via a weighted formula to amplify variance. For example, if for the 250 Hz frequency a dock was given a score of 4 on 10, it meant that it was able to reproduce the tone but with accompanying distortion. In this case it got a final score of $4 \times 1 = 4$. If it got 5 then the final score was 5×2 , and so on.

The next set of samples tested the qualitative aspects of performance. Each track tested

a particular aspect of audio tonality such as stereo separation, vocal performance, instrument separation, and detail etc. Tracks were also selected based on different genre:

- A classical sample with a 200 year old string bass to check string resonance.
- Bass reproduction on Massive Attack's **bass** heavy track Angel for accuracy of lows.
- Chris Rea's track *Call On Me* to check if speakers retained that husky edge in **vocals**.
- **Stereo separation** was tested on an old favourite: *Whole Lotta Love* - Led Zeppelin.
- **Binaural Stereo tonality** sample had two instruments in different pitch but in harmony
- The operatic orchestration on Therion's *O Fortuna* was used to judge **highs**.
- We used an edited version of *Selkies* by Between The Buried and Me to check for **instrument separation**.
- Dream Theater's *Pull Me Under* was used to gauge a **mixture** of performance areas

same as *Time* by Pink Floyd for **detail**, which is in itself a masterpiece of production.

- **Percussion** on *Brutal Hearts* by Bedouin Soundclash is good enough to literally shake up any driver.

Test tracks were either fully uncompressed .wav files, 320 kbps MP3 or Apple Lossless. Refer to the table to see what weights we assigned to the frequency and qualitative tests, design and build, and volume.

Features: Every feature was assigned a weight. The features included fairly essential ones as Stereo playback and different auxiliary sources, to bonus features like iPad docking. Stereo was given a weight of 20 per cent, same as the ability to connect via USB and play music off the PC. In the final scoring, Performance accounted for 80 percent, the rest contributed by Features.

the ability to set multiple alarms via the custom app designed by Philips. The Philips app which works with all of their Docks turned out to be the best app in the test. It lets you set multiple alarms, lets you wake up to music on your iPod, and even some nature sounds (like ocean waves, birds, etc) which come with the app. The app can stream tracks from TuneIn Radio, which features a few local channels. It does not have an in-built FM tuner so this is a small way in which the absence can be circumvented. The app does have its flaws though. It does not allow you to wake up to an FM station. Not much can be expected in terms of performance at this price point. There isn't much bass to speak of and the last 5 seconds of our 17Hz to 20Hz sweep were blank. Some frequencies around 500 Hz were distorted. The 4W speakers will fill a small room with sound and just about enough background music for your morning wake up routine.

The bass line on *Call On Me* literally obliterated the diaphragm on what must imaginably be really tiny drivers. Whatever customisations Philips has made for Omni directional sound has made stereo separation almost non-existent.

Logitech Bedside Dock

In terms of looks this is one tacky device. The build quality is also lackluster with buttons resembling the kind you'd find on VCRs twenty years ago. The only performance area it scores above the Philips Fidelio 1100 is stereo separation. Otherwise distortion pretty much began from the upper highs to the pure mids. Binaural stereo tonality test caused reverberation throughout. It's almost as if something inside the casing resonates at multiple frequencies - that bad. There's a closed bass reflex port at the back but it does little to increase the bass. The device gets feature points for an LCD display, a hardware

(non-app-based) alarm that works irrespective of external power thanks to two AAA batteries, and an FM radio tuner built into the device.

F&D i224

The i224 docking station's triangular shape does catch your eye. It sounds just as good (or bad) as the Logitech as long as you don't take the volume past the two-third mark. The Logitech seems to have slightly larger drivers and is better with percussions. But this little guy is loud. If you want your dock to make a noise and make its presence felt, the F&D i224 won't disappoint.

Category verdict: The F&D is exceptionally cheap and hence

managed to tip the scales of value in its favour, winning our Best Buy for this category. However, go for it only if you primarily need a loud iPod dock. The Logitech Bedside Dock should be avoided at all cost as being able to dock an iPad can never be reason enough to justify that purchase. The Philips Fidelio 1100 is the best performing dock in this category and won us over with its warm sound signature and good looks.

Portable docks

Our motley crew of iPod Docks had three that were a little different from the rest. They came with carrying sleeves or built in handles, were portable and worked on battery. They either had built-in lithium ion / nickel metal hydride batteries or worked on AA batteries. Let's look at the three in detail.

Yamaha PDX-11

This is one strange piece of audio equipment. Strange for a couple of reasons - it's not stereo, it



F&D i224